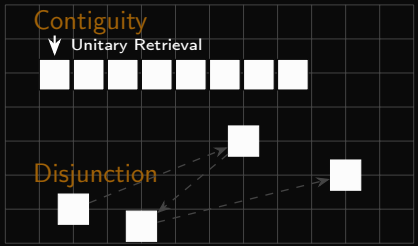
Why Algorithms Need the Machine

Locality, Speed, and the 16-Structure Sprint

Sheikh Mohsin

Systems Programming | April 2026 | Prof. Sanjay Patel

The Spatial Premise



- Modern CPUs operate on a **spatial premise**.
- Contiguous layouts leverage hardware prefetching.
- Pointer-chasing induces serial stalls.
- Memory latency is the **primary bottleneck** of the 21st-century.

The Scale of the Project

16
Data Structural Families

6,641
Datapoints

730B
Memory Elements Analyzed

The Reality: Locality is a Multiplier on Big-O

Big-O defines the *curve*. Hardware locality defines the *crossover point*.

> *"Memory Access is like a Coffee Shop vs. a Warehouse"*

L1 Cache (Coffee Shop)
Right in front of you. Fast. Automatic.

RAM (The Warehouse)
Slow. Requires a trip. CPU stalls while waiting.

For operations with identical complexity, spatial locality dictates the winner.

Meet the Contenders: Beyond the BST

How do we organize data when a simple list isn't enough?

SkipList

The Express Elevator

A list with "shortcut" lanes to skip nodes. Fast as a Tree, but simpler.

vEB Tree

Digital Fractals

Uses recursion to divide data into clusters. Theoretically the fastest.

B-Tree

The Librarian

Stores hundreds of items per node. Keeps the CPU happy by minimizing trips.

Measured all 16 contenders across 5 orders of magnitude.

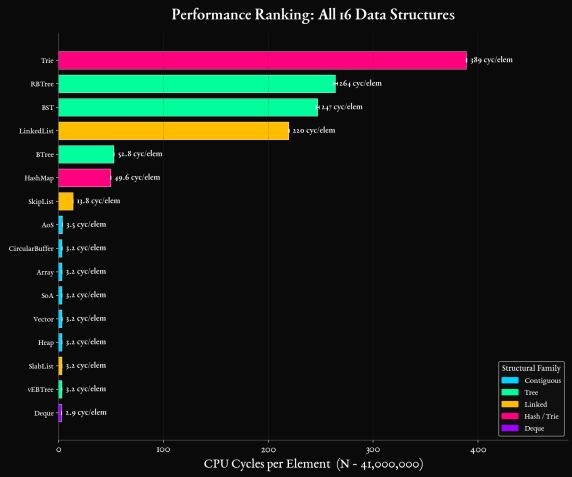
Methodology: The Precision Toolkit

```
// CPU Pinning
taskset -c 1 ./benchmark

// Memory Barrier
__asm__ __volatile__("" ::: "
    memory");

// Hardware PMU Access
syscall(SYS_perf_event_open,
    ...);

// Optimization Shield
volatile long long sink = 0;
```

The Scientific Rigor:

- **taskset:** Core isolation.
- **asm/volatile:** Prevents compiler 'magic'.
- **perf:** True cycle counts.

Who Wins the Sprint? (N = 41 Million)

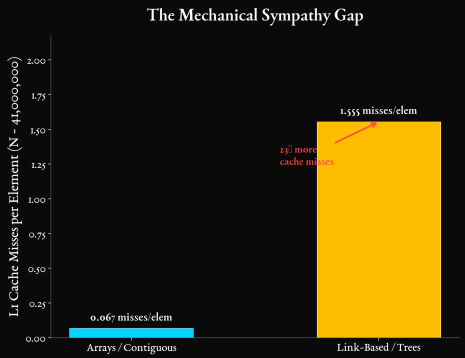


The Traversal Bound

Arrays win by 40x *during full scans.*

But remember: inserting into an array costs $O(N)$ data movement.

The Secret Multiplier: Locality



> *"Locality is like Shopping in one aisle vs. running across the store."*

Pointer Tax: Links or Pointers force the CPU to find a new address for every single element.

Practical Lessons

Algorithms are Math, but Performance is Physics.

I. **Contiguity is King (for Retrieval):** Locality maximizes throughput.

II. **The Write Penalty:** Beware $O(N)$ data movement during mutations.

III. **The B-Tree Compromise:** Balance Big-O efficiency with cache contiguity.

The Big Three Bottlenecks

1. The Cache Wall

0.90 Correlation

Cycles track LLC misses perfectly. If you miss, the CPU stops for 300 cycles.

2. The Translation Wall

Page Table Overload

Pointer structures trigger TLB misses, forcing multi-cycle translations.

3. The Structural Tax

402x Latency Delta

Random memory access kills the hardware prefetcher. $O(\log N)$ pointers lose to $O(N)$ scanning.

> *"Bottlenecks is like Traffic Jams vs. Open Highways"*

Any Questions?

Thank You!

Sheikh Mohsin

Systems Programming | 2026

SheikhMohsin9311/Performance-and-Data-Represent

Technical Documentation & Data